

JESS Thermodynamic Database v8.9

Hafnium

24-Jan-23

Reaction No. 78							
Hf+4 + H2O = Hf+4_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.25 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-2.5 (2SF)	0	CRV	23624
3	25	0	Inf. Dilution	lgK:-0.29 (2SF)	4	CRV	32224
4	25	0.7	Seawater	lgK:-0.79 (2SF)	0	ENS	6496
5	25	4	NaClO4	lgK:0.08 (1SF)	0	MLC	403
6	25	0	Inf. Dilution	dH:10.3 (3SF)	0	CRV	6478
7	25	0.7	Seawater	dH:10.3 (3SF)	0	ENS	6496

Reaction No. 2745							
Hf+4 + H+1_F-1 = Hf+4_F-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.2 (2SF)	1	ELC	
2	25	3	NaClO4	lgK:4.42 (3SF)	0	MGL	5398
3	20	4	HClO4	lgR:5.5 (0.07SD)	0	CMN	437
4	25	3	HClO4	lgR:4.16 (3SF)	0	MDS	931
5	25	3	HClO4	lgR:4.42 (3SF)	0	MGL	5398
6	25	4	HClO4	lgR:5.5 (2SF)	0	MIS	5077
7	27	2	HClO4	lgR:4.62 (3SF)	0	MDS	931

Reaction No. 2746							
Hf+4_F-1 + H+1_F-1 = Hf+4_F-1 (2) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.6 (2SF)	1	ELC	
2	25	3	NaClO4	lgK:3.12 (3SF)	0	MGL	5398
3	20	4	HClO4	lgR:4.0 (0.07SD)	0	CMN	437
4	25	3	HClO4	lgR:3.12 (3SF)	0	MGL	5398
5	25	4	HClO4	lgR:4.01 (3SF)	0	MIS	5077

Reaction No. 2747							
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Hf+4_F-1(2) + H+1_F-1 = Hf+4_F-1(3) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.6(2SF)	1	ELC	
2	25	3	NaClO4	lgK:2.64(3SF)	0	MGL	5398
3	20	4	HClO4	lgR:3.0(0.07SD)	0	CMN	437
4	25	3	HClO4	lgR:2.64(3SF)	0	MGL	5398

Reaction No. 2748							
Hf+4_F-1(3) + H+1_F-1 = Hf+4_F-1(4) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.4(2SF)	1	ELC	
2	25	3	NaClO4	lgK:1.9(2SF)	0	MGL	5398
3	20	4	HClO4	lgR:2.2(0.07SD)	0	CMN	437
4	25	3	HClO4	lgR:1.9(2SF)	0	MGL	5398

Reaction No. 2749							
Hf+4_F-1(4) + H+1_F-1 = Hf+4_F-1(5) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.6(2SF)	1	ELC	
2	25	3	NaClO4	lgK:1.59(3SF)	0	MGL	5398
3	20	4	HClO4	lgR:1.7(0.07SD)	0	CMN	437
4	25	3	HClO4	lgR:1.59(3SF)	0	MGL	5398

Reaction No. 2750							
Hf+4_F-1(5) + H+1_F-1 = Hf+4_F-1(6) + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	0	EDH	
2	25	0	Inf. Dilution	lgK:-5.4(2SF)	1	ELC	
3	25	3	NaClO4	lgK:1.70(3SF)	0	MGL	5398
4	20	4	HClO4	lgR:0.5(0.07SD)	0	CMN	437
5	25	3	HClO4	lgR:1.70(3SF)	0	MGL	5398

Reaction No. 3083							
NTA-3 + Hf+4 = Hf+4_NTA-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.23	HClO4	lgK:20.342(5SF)	3	MIX	4405
2	25	0	Inf. Dilution	lgK:23.6(3SF)	4	CRV	552

3	37	0.15	NaClO4	lgK:17.30 (0.175SD)	0	MGL	2151
Note (3): Low wgt for internal consistency							
4	25	0	Inf. Dilution	dH:4.9 (2SF)	4	CRV	552

Reaction No. 4195							
EDTA-4 + Hf+4 = Hf+4_EDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.23	HClO4	lgK:29.491 (5SF)	4	MIX	4405
2	25	0	Inf. Dilution	lgK:33.61 (4SF)	1	EOR	
3	25	0	Inf. Dilution	lgK:33.7 (3SF)	4	CRV	552
4	25	0.5	KCl	lgK:15.1 (3SF)	0	MSP	11516
Note (4): Ternovaya T et al., Zhur. Neorg. Khim., 1978, 23, 1215							
5	25	0	Inf. Dilution	dH:-0.2 (1SF)	4	CRV	552

Reaction No. 4271							
DTPA-5 + Hf+4 = Hf+4_DTPA-5							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.23	HClO4	lgK:35.398 (5SF)	0	MIX	4405
2	25	0	Inf. Dilution	lgK:35.9 (3SF)	1	ESC	

Reaction No. 4370							
Hf+4 + H+1(2)_TTHA-6 = Hf+4_H+1(2)_TTHA-6							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.5	Unknown	lgK:19.1 (3SF)	4	ENS	773

Reaction No. 4603							
Kojic-1 + Hf+4 = Hf+4_Kojic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:12.04 (4SF)	6	MDS	999
2	20	2	HClO4	lgK:12.20 (4SF)	6	MDS	999

Reaction No. 4604							
2<Kojic-1> + Hf+4 = Hf+4_Kojic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:22.59 (4SF)	6	MDS	999

Reaction No. 4788							
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Hf+4 + 4<e-1> = Hf(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:-107.(3SF)	0	ENS	140
2	25	0	Inf. Dilution	dG:555.(3SF) kJ	0	CRV	7421
3	25	0	Inf. Dilution	dG:132.600 (6SF)	0	CRV	9029
4	25	0	Inf. Dilution	E:-1.55 (3SF)	3	CRV	6330
5	25	0	Inf. Dilution	E:-1.55 (3SF)	3	CRV	7761
6	25	0	Inf. Dilution	E:-1.7 (2SF)	0	CRV	22551
7	25	0	Inf. Dilution	dH:676.5 (4SF) kJ	3	CRV	7761
8	25	0	Inf. Dilution	dH:150.250 (6SF)	0	CRV	9029

Reaction No. 4816							
HfO2(s) + 4<e-1> + 4<H+1> = Hf(s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.57 (0.01SD)	3	MDG	775
2	25	0	Inf. Dilution	E:-1.505 (4SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-1.591 (4SF)	2	CRV	7761
4	25	0	Inf. Dilution	E:-1.57 (3SF)	5	CRV	22551
5	25	0	Inf. Dilution	dH:573.2 (4SF) kJ	3	CRV	7761

Reaction No. 9198							
Hf+4 + H+1(3)_NTA-3 = Hf+4_NTA-3 + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:5.1 (0.09SD)	4	MIX	1118
2	25	2	HClO4	lgR:3.83 (0.02SD)	4	MIX	1118

Reaction No. 11849							
Maltol-1 + Hf+4 = Hf+4_Maltol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:13.16 (4SF)	6	MDS	999
2	20	2	HClO4	lgK:13.24 (4SF)	6	MDS	999

Reaction No. 11850							
2<Maltol-1> + Hf+4 = Hf+4_Maltol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:24.48 (4SF)	6	MDS	999
2	20	2	HClO4	lgK:24.18 (4SF)	6	MDS	999

Reaction No. 12748							
HIMDA-2 + Hf+4 = Hf+4_HIMDA-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.23	HClO4	lgK:14.63 (4SF)	0	MIX	931
2	25	0.1	Unknown	lgK:8.4 (0.2SD)	2	CRV	552
3	35	0.1	KNO3	lgK:9.7 (2SF)	2	MGL	2261
4	25	0.1	Unknown	dH:-2.6 (2SF)	2	CRV	552
5	25	0.5	Unknown	dH:-3.2 (2SF)	2	CRV	552
6	25	1	Unknown	dH:-3.4 (2SF)	2	CRV	552

Reaction No. 13679							
2ThenTriFAcone-1 + Hf+4 = Hf+4_2ThenTriFAcone-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:10.60 (4SF)	6	MDS	999

Reaction No. 13680							
2<2ThenTriFAcone-1> + Hf+4 = Hf+4_2ThenTriFAcone-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:21.44 (4SF)	6	MDS	999

Reaction No. 13681							
3<2ThenTriFAcone-1> + Hf+4 = Hf+4_2ThenTriFAcone-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:31.50 (4SF)	6	MDS	999

Reaction No. 13682							
4<2ThenTriFAcone-1> + Hf+4 = Hf+4_2ThenTriFAcone-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:41.52 (4SF)	6	MDS	999

Reaction No. 18067							
Hf+4_EDTA-4 + H2O = Hf+4_OH-1_EDTA-4 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:-6.00 (3SF)	6	ENS	1539
2	25	0	Inf. Dilution	lgK:-5.8 (2SF)	1	ESC	

Reaction No. 18171							
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Hf+4 + H+1(2)_Citric-3 = Hf+4_H+1(2)_Citric-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:5.60(3SF)	6	ENS	1539

Reaction No. 18243							
Oxalic-2 + Hf+4 = Hf+4_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:10.10(4SF)	6	ENS	1539
2	20	2	Na(ClO4)	lgK:10.22(4SF)	5	MDS	906
3	25	0	Inf. Dilution	lgK:9.913(4SF)	1	ELC	

Reaction No. 18244							
2<Oxalic-2> + Hf+4 = Hf+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:19.40(4SF)	0	ENS	1539
2	20	2	Na(ClO4)	lgK:18.88(4SF)	5	MDS	906
3	25	0	Inf. Dilution	lgK:20.2(3SF)	1	ELC	

Reaction No. 18310							
Tartaric-2 + Hf+4 = Hf+4_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:5.20(3SF)	6	ENS	1539
2	25	2	NaClO4	lgK:4.68(3SF)	5	MDS	13484

Reaction No. 18311							
Tiron-4 + Hf+4 = Hf+4_Tiron-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:23.00(4SF)	6	ENS	1539
2	23	0.19	HClO4	lgK:24.663(5SF)	0	MIX	4452
Note (2): Low wgt for internal consistency							

Reaction No. 18312							
2<Tiron-4> + Hf+4 = Hf+4_Tiron-4(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:44.92(4SF)	6	ENS	1539

Reaction No. 18313							
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3<Tiron-4> + Hf+4 = Hf+4_Tiron-4(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HClO4	lgK:66.92(4SF)	3	MIX	4452

Reaction No. 18526							
Hf+4 + H+1_Gly-1 = Hf+4_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	lgK:1.5(0.07SD)	3	MKN	1555

Reaction No. 27187							
Hf+4_OH-1_IDA-2 + H+1 = Hf+4_IDA-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	35	0.1	KNO3	lgK:10.9(0.2SD)	3	MGL	1889

Reaction No. 27188							
2<Hf+4_EDTA-4> + 2<H2O> = Hf+4(2)_OH-1(2)_EDTA-4(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.8(2SF)	1	ESC	
2	35	0.1	KNO3	lgK:-9.00(3SF)	3	MGL	1889

Reaction No. 27528							
2<Hf+4_HOEDTA-3> + 2<H2O> = Hf+4(2)_OH-1(2)_HOEDTA-3(2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.0(3SF)	1	ESC	
2	35	0.1	KNO3	lgK:-16.15(4SF)	3	MGL	1889

Reaction No. 27529							
Hf+4_OH-1_HIMDA-2 + H+1 = Hf+4_HIMDA-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:9.44(3SF)	3	CRV	552
2	25	0	Inf. Dilution	lgK:9.0(2SF)	1	EES	
3	35	0.1	KNO3	lgK:9.7(0.2SD)	0	MGL	1889
4	35	0.1	KNO3	lgK:8.65(3SF)	3	MGL	2261

Reaction No. 32311							
Hf+4_OH-1 + H+1_Mucic-2 = Hf+4_Mucic-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	1	KCl	lgK:9.68 (3SF)	5	MSP	2261
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Reaction No. 37306							
Salicylaldoxime-2 + Hf+4 = Hf+4_Salicylaldoxime-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.7 (3SF)	0	MGL	931
Note (1): Low wgt for internal consistency							
2	25	0.01	KCl	lgK:15.82 (4SF)	0	MGL	931
3	25	0.025	KCl	lgK:14.15 (4SF)	0	MGL	931
4	25	0.05	KCl	lgK:13.00 (4SF)	0	MGL	931
5	25	0.075	KCl	lgK:12.30 (4SF)	0	MGL	931
6	25	0.1	KCl	lgK:11.05 (4SF)	3	MGL	931

Reaction No. 38849							
Hf+4_OH-1(3) + Acetic-1 = Hf+4_Acetic-1_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:3.03 (3SF)	4	MHY	906

Reaction No. 40128							
Hf+4_OH-1 + H+1(2)_Oxalic-2 = Hf+4_Oxalic-2 + H+1 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.3 (2SF)	1	ESC	
2	25	4	NaClO4	lgK:4.398 (4SF)	4	MDS	4708

Reaction No. 40383							
Hf+4 + H+1(2)_Oxalic-2 = Hf+4_Oxalic-2 + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.1 (2SF)	1	ELC	
2	23	2	HClO4	lgR:5.2 (0.1SD)	3	MIX	931
3	23	4	HClO4	lgR:5.2 (0.1SD)	3	MIX	931

Reaction No. 40384							
Hf+4 + 2<H+1(2)_Oxalic-2> = Hf+4_Oxalic-2(2) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.1 (2SF)	1	ELC	
2	23	2	HClO4	lgR:9.7 (0.1SD)	3	MIX	931

Reaction No. 40911							
$\text{Hf}+4 + \text{H}+1(2)\text{_}234\text{TriOHPentDioic-2} = \text{Hf}+4\text{_H}+1\text{_}234\text{TriOHPentDioic-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.42(3SF)	5	MIX	931
2	23	2	HClO4	lgR:2.7(0.03SD)	5	MIX	3262

Reaction No. 40912							
$\text{Hf}+4 + 2\langle\text{H}+1(2)\text{_}234\text{TriOHPentDioic-2}\rangle = \text{Hf}+4\text{_H}+1(2)\text{_}234\text{TriOHPentDioic-2}(2) + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:4.9(2SF)	5	MIX	931
2	23	2	HClO4	lgR:5.2(0.1SD)	5	MIX	3262

Reaction No. 41441							
$\text{IDA-2} + \text{Hf}+4 = \text{Hf}+4\text{_IDA-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	35	0.1	KNO3	lgK:10.9(3SF)w	5	MPH	2261

Reaction No. 41689							
$\text{Hf}+4\text{_OH-1}(3) + \text{H}+1(3)\text{_Pyrogallol-3} = \text{Hf}+4\text{_H}+1\text{_OH-1}(2)\text{_Pyrogallol-3} + \text{H}+1 + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	18:20	0	Inf. Dilution	lgK:4.44(3SF)	4	MSP	4406
2	18:20	0.1	KCl	lgK:4.33(3SF)	4	MSP	4406

Reaction No. 42043							
$\text{Hf}+4\text{_OH-1}(4) + \text{H}+1(2)\text{_Tiron-4} + 2\langle\text{H}+1\rangle = \text{Hf}+4\text{_Tiron-4} + 4\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.2	HClO4	lgR:4.398(4SF)	3	MIX	4452

Reaction No. 42250							
$\text{Hf}+4 + \text{H}+1(2)\text{_Lactic-1} = \text{Hf}+4\text{_H}+1\text{_Lactic-1} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.03(3SF)	5	MIX	3262

Reaction No. 42251							
$\text{Hf}+4 + 2\langle\text{H}+1(2)\text{_Lactic-1}\rangle = \text{Hf}+4\text{_H}+1(2)\text{_Lactic-1}(2) + 2\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.03(3SF)	5	MIX	3262

Reaction No. 42254							
$\text{Hf}+4 + \text{H}+1(2)\text{_Malic-2} = \text{Hf}+4\text{_H}+1\text{_Malic-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:1.83 (3SF)	5	MIX	3262

Reaction No. 42256							
$\text{Hf}+4 + \text{H}+1(2)\text{_Tartaric-2} = \text{Hf}+4\text{_H}+1\text{_Tartaric-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:1.99 (3SF)	5	MIX	3262

Reaction No. 42258							
$\text{Hf}+4 + \text{H}+1(2)\text{_Citric-3} = \text{Hf}+4\text{_H}+1\text{_Citric-3} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	HClO4	lgR:2.54 (3SF)	5	MIX	3262

Reaction No. 42379							
$\text{Hf}+4 + \text{H}+1(2)\text{_Oxalic-2} = \text{Hf}+4\text{_H}+1\text{_Oxalic-2} + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.2 (2SF)	1	ESC	
2	23	2	HClO4	lgR:5.185 (4SF)	4	MDG	3262

Reaction No. 42380							
$\text{Hf}+4 + 2\text{<H}+1(2)\text{_Oxalic-2>} = \text{Hf}+4\text{_H}+1(2)\text{_Oxalic-2(2)} + 2\text{<H}+1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.5 (2SF)	1	ESC	
2	23	2	HClO4	lgR:8.535 (4SF)	4	MDG	3262

Reaction No. 42381							
$\text{Hf}+4 + 3\text{<H}+1(2)\text{_Oxalic-2>} = \text{Hf}+4\text{_H}+1(3)\text{_Oxalic-2(3)} + 3\text{<H}+1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.1 (3SF)	1	ESC	
2	23	2	HClO4	lgR:10.19 (4SF)	4	MIX	3262

Reaction No. 42382							
$\text{Hf}+4 + 4\text{<H}+1(2)\text{_Oxalic-2>} = \text{Hf}+4\text{_H}+1(4)\text{_Oxalic-2(4)} + 4\text{<H}+1\text{>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.7 (3SF)	1	ESC	

2	23	2	HClO4	lgR:13.89(4SF)	4	MIX	3262
3	23		MeCN	lgK:2.66(0.01SD)	5	EOD	2875

Reaction No. 42898							
Hf+4_Acetic-1_OH-1(3) + Acetic-1 = Hf+4_Acetic-1(2)_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.01	Unknown	lgK:1.90(3SF)	4	MHY	906

Reaction No. 42899							
Hf+4_OH-1(2) + Acetic-1 = Hf+4_Acetic-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.7(0.05SD)	4	MKN	906

Reaction No. 42945							
Hf+4_OH-1(2) + Glycolic-1 = Hf+4_Glycolic-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.3(2SF)	4	MKN	906
2	25	0.01	Unknown	lgK:7.1(2SF)	4	MKN	906
3	25	0.1	Unknown	lgK:7.3(2SF)	4	MKN	906

Reaction No. 43150							
Hf+4_OH-1(3) + H+1_Ser-1 = Hf+4_H+1_OH-1(3)_Ser-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:1.3(0.06SD)	4	MKN	906

Reaction No. 43246							
Hf+4_OH-1(3) + Tartaric-2 = Hf+4_OH-1(3)_Tartaric-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.03(3SF)	4	MHY	906

Reaction No. 43577							
Hf+4_OH-1(3) + 234TriOHPentDioic-2 = Hf+4_OH-1(3)_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:6.50(3SF)	4	MHY	906

Reaction No. 43701							
Hf+4 + H+1_Ascorbic-2 = Hf+4_H+1_Ascorbic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	23			lgK:4.2(2SF)	3	MSP	906
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Reaction No. 44182							
Hf+4_OH-1(3) + H+1_Salicylic-2 = Hf+4_H+1_OH-1(3)_Salicylic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			lgK:5.5(0.2SD)	4	MKN	906

Reaction No. 44340							
Hf+4_OH-1(3) + H+1_SulfSal-3 = Hf+4_H+1_OH-1(3)_SulfSal-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.5(2SF)	1	ESC	
2	25			lgK:6.5(0.2SD)	0	MKN	906

Reaction No. 44468							
Hf+4_OH-1(2) + 2<PhenAcet-1> = Hf+4_OH-1(2)_PhenAcet-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.0(2SF)	4	MKN	906

Reaction No. 45084							
TriMDTA-4 + Hf+4 = Hf+4_TriMDTA-4							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:27.65(4SF)	4	MDS	906

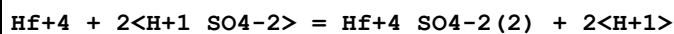
Reaction No. 45085							
Hf+4 + 2<H+1_TriMDTA-4> = Hf+4_H+1(2)_TriMDTA-4(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HClO4	lgK:54.43(4SF)	4	MDS	906

Reaction No. 53162							
Hf+4_F-1(5) + F-1 = Hf+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:3.83(3SF)	3	MIX	5398
2	25	0	Inf. Dilution	lgK:-2.3(2SF)	1	ELC	

Reaction No. 53307							
Hf+4 + H+1_SO4-2 = Hf+4_SO4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2.3	HClO4	lgR:2.11(3SF)	4	MCE	140

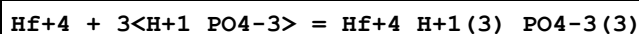
2	25	1	HClO4	lgR:1.9 (2SF)	5	MDS	5398
3	27	2	HClO4	lgR:2.04 (3SF)	5	MDS	931

Reaction No. 53308



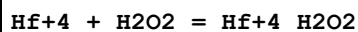
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2.3	HClO4	lgR:3.32 (3SF)	5	MCE	931
2	25	1	HClO4	lgR:2.8 (2SF)	5	MDS	5398
3	27	2	HClO4	lgR:3.7 (2SF)	5	MDS	931

Reaction No. 53371



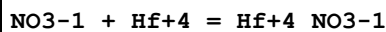
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	NaClO4	lgK:1.57 (3SF)	5	MSP	5398

Reaction No. 53505



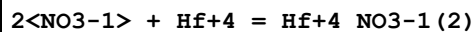
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23			lgK:5.1 (2SF)	0	MKN	5398

Reaction No. 53658



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.15 (2SF)	5	MCE	931
2	20	4	HClO4	lgK:0.92 (2SF)	0	MDS	140
3	20	4	Unknown	lgK:0.40 (2SF)	0	CRV	552
4	23	2	HClO4	lgK:-0.05 (1SF)	4	MCE	140
5	23	4	HClO4	lgK:-0.22 (2SF)	4	MCE	140
6	23	4	HClO4	lgK:-0.4 (1SF)	5	MDS	5398
7	25	0	Inf. Dilution	lgK:-0.2005 (4SF)	1	ELC	
8	25	2	HClO4	lgK:0.34 (2SF)	0	MDS	140
9	25	2	Unknown	lgK:0.34 (2SF)	0	CRV	552
10	27	2	HClO4	lgK:-0.1 (1SF)	5	MDS	931

Reaction No. 53659



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	20	2	HClO4	lgK:-0.96 (2SF)	0	MCE	931
2	20	4	HClO4	lgK:1.51 (3SF)	0	MDS	140
3	20	4	Unknown	lgK:0.40 (2SF)	0	CRV	552
4	23	2	HClO4	lgK:-0.26 (2SF)	0	MCE	140
5	23	4	HClO4	lgK:-0.92 (2SF)	0	MCE	140
6	23	4	HClO4	lgK:-1.3 (2SF)	0	MDS	5398
7	25	0	Inf. Dilution	lgK:0.1941 (4SF)	1	EOR	
8	25	2	HClO4	lgK:-1.52 (3SF)	4	MDS	140
9	25	2	Unknown	lgK:0.0 (1SF)	0	CRV	552

Reaction No. 54051							
$\text{Hf}+4 + 3\langle\text{H}+1_ \text{SO}_4-2\rangle = \text{Hf}+4_ \text{SO}_4-2(3) + 3\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2.3	HClO4	lgR:6.48 (3SF)	3	MCE	931

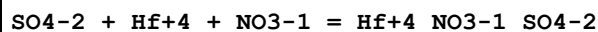
Reaction No. 54052							
$\text{SO}_4-2 + \text{Hf}+4 = \text{Hf}+4_ \text{SO}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:3.10 (3SF)	5	MCE	931
2	25	0	Inf. Dilution	lgK:8.482 (4SF)	1	EOR	
3	25	2	HClO4	lgK:3.11 (3SF)	5	MDS	140
4	25	2	Unknown	lgK:3.04 (3SF)	6	CRV	552
5	25	4	NaClO4	lgK:1.00 (3SF)	5	MDS	4708

Reaction No. 54053							
$2\langle\text{SO}_4-2\rangle + \text{Hf}+4 = \text{Hf}+4_ \text{SO}_4-2(2)$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:5.42 (3SF)	5	MCE	931
2	25	0	Inf. Dilution	lgK:13.56 (4SF)	1	EOR	
3	25	2	NaClO4	lgK:5.48 (3SF)	5	MDS	11516
Note (3): Peshkova V et al., Vestnik Moskov Univ., 1963, 18, 40							
4	25	2	Unknown	lgK:5.44 (3SF)	6	CRV	552
5	25	4	NaClO4	lgK:2.26 (3SF)	5	MDS	4708

Reaction No. 54054							
$\text{SO}_4-2 + \text{Hf}+4 + \text{Cl}-1 = \text{Hf}+4_ \text{Cl}-1_ \text{SO}_4-2$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

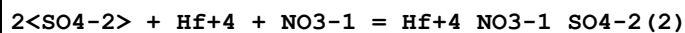
1	20	2	HClO4	lgK:3.04 (3SF)	5	MCE	931
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Reaction No. 54055



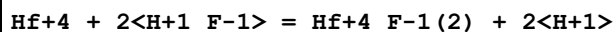
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:3.23 (3SF)	5	MCE	931

Reaction No. 54056



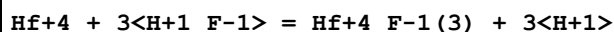
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:5.7 (2SF)	5	MCE	931

Reaction No. 54117



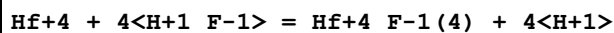
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:7.79 (3SF)	5	MDS	931

Reaction No. 54118



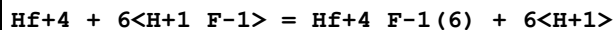
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:10.10 (4SF)	5	MDS	931

Reaction No. 54119



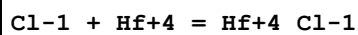
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:12.04 (4SF)	5	MDS	931

Reaction No. 54120



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:12.62 (4SF)	5	MDS	931

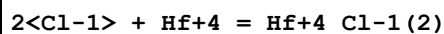
Reaction No. 54172



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:0.07 (1SF)	5	MCE	931
2	20	3	HClO4	lgK:0.18 (2SF)	5	MCE	931
3	20	3	HClO4	lgK:0.34 (2SF)	5	MDS	931

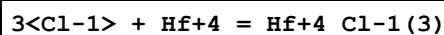
4	23	2	H(ClO4)	lgK:-0.02 (1SF)	3	MCE	140
5	23	4	H(ClO4)	lgK:-0.03 (1SF)	3	MCE	140
6	25	2	H(ClO4)	lgK:0.38 (2SF)	4	MDS	140
7	27	2	HClO4	lgK:-0.15 (2SF)	5	MDS	931

Reaction No. 54173



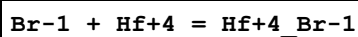
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.48 (2SF)	5	MCE	931
2	20	3	HClO4	lgK:-0.02 (1SF)	0	MDS	931
3	23	2	H(ClO4)	lgK:-0.92 (2SF)	0	MCE	140
4	23	4	H(ClO4)	lgK:-0.74 (2SF)	3	MCE	140
5	25	0	Inf. Dilution	lgK:2.7 (2SF)	1	EDH	
6	27	2	HClO4	lgK:-0.32 (2SF)	5	MDS	931

Reaction No. 54174



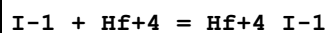
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	HClO4	lgK:-0.40 (2SF)	1	MCE	931
2	23	2	H(ClO4)	lgK:-1.15 (3SF)	0	MCE	140
3	23	4	H(ClO4)	lgK:-1.10 (3SF)	0	MCE	140
4	25	0	Inf. Dilution	lgK:1.2 (2SF)	1	EES	

Reaction No. 54285



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	HClO4	lgK:-0.1 (1SF)	4	MDS	931

Reaction No. 54345



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	3	HClO4	lgK:-0.46 (2SF)	4	MDS	931
2	20	3	Unknown	lgK:-0.5 (1SF)	3	CRV	7271
3	25	0	Inf. Dilution	lgK:-0.5 (1SF)	1	EES	
4	25	0	Inf. Dilution	lgK:-0.5 (4SF)	1	EOR	

Reaction No. 54497

Hf+4_F-1(4) + F-1 = Hf+4_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25			dH:2.988(4SF)	3	MCL	437

Reaction No. 58385							
3<NO3-1> + Hf+4 = Hf+4_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:1.89(3SF)	0	MDS	140
2	25	0	Inf. Dilution	lgK:-0.03612(4SF)	1	EOR	
3	25	2	Unknown	lgK:-0.7(1SF)	3	CRV	552

Reaction No. 58386							
4<NO3-1> + Hf+4 = Hf+4_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:2.08(3SF)	0	MDS	140
2	25	0	Inf. Dilution	lgK:-0.5912(4SF)	1	ELC	

Reaction No. 58387							
5<NO3-1> + Hf+4 = Hf+4_NO3-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:2.08(3SF)	0	MDS	140

Reaction No. 58388							
6<NO3-1> + Hf+4 = Hf+4_NO3-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	4	HClO4	lgK:1.81(3SF)	0	MDS	140

Reaction No. 58389							
Hf+4_NO3-1 + NO3-1 = Hf+4_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	HClO4	lgK:-0.34(2SF)	4	MDS	140

Reaction No. 58390							
Hf+4_NO3-1(2) + NO3-1 = Hf+4_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	HClO4	lgK:-0.72(2SF)	4	MDS	140

Reaction No. 58391							
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Hf+4_NO3-1(3) + NO3-1 = Hf+4_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	HClO4	lgK:-0.80(2SF)	4	MDS	140

Reaction No. 58513							
Hf+4_SO4-2 + SO4-2 = Hf+4_SO4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO4	lgK:2.32(3SF)	5	MIX	11516
Note (1): Ermakov A et al., Zhur. Neorg. Khim., 1967, 12, 3372							
2	25	2	HClO4	lgK:2.47(3SF)	5	MDS	140
3	25	2	NaClO4	lgK:2.37(3SF)	5	MDS	11516
Note (3): Peshkova V et al., Vestnik Moskov Univ., 1963, 18, 40							
4	25	4	NaClO4	lgK:1.26(3SF)	5	MDS	4708

Reaction No. 58577							
4<Cl-1> + Hf+4 = Hf+4_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	2	H(ClO4)	lgK:-1.10(3SF)	3	MCE	140
2	25	2	H(ClO4)	lgK:-1.31(3SF)	4	MDS	140

Reaction No. 58578							
Hf+4_Cl-1 + Cl-1 = Hf+4_Cl-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.1(2SF)	1	ELC	
2	25	2	H(ClO4)	lgK:-0.31(2SF)	4	MDS	140

Reaction No. 58579							
Hf+4_Cl-1(2) + Cl-1 = Hf+4_Cl-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.0(2SF)	1	ELC	
2	25	2	H(ClO4)	lgK:-0.68(2SF)	2	MDS	140

Reaction No. 58580							
Hf+4_Cl-1(3) + Cl-1 = Hf+4_Cl-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(1SF)	1	EES	
2	25	2	H(ClO4)	lgK:-0.7(1SF)	2	MDS	140

Reaction No. 60207							
Imidazole + Hf+4 = Hf+4_Imidazole							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.5	NaClO4	dH:-7.95(3SF)	6	CRV	5406
2	25	0.5	Unknown	dH:-7.86(3SF)	6	CRV	5406

Reaction No. 60504							
OH-1 + Hf+4 = Hf+4_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.75(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:13.7(3SF)	4	CRV	2556
3	25	0	Inf. Dilution	lgK:13.8(3SF)	4	MSL	14521
4	25	4	Unknown	lgK:13.3(3SF)	5	CRV	2556

Reaction No. 60505							
5<OH-1> + Hf+4 = Hf+4_OH-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.7(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:52.8(3SF)	0	CRV	2556
3	25	0	Inf. Dilution	lgK:49.7(0.2SD)	4	MSL	14521

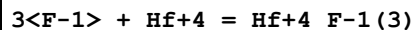
Reaction No. 60506							
HfO2(s) + 2<H2O> = Hf+4 + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-57.5(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-54.8(3SF)	0	CRV	2556
3	25	1	Unknown	lgK:-52.7(3SF)	2	ENS	773

Reaction No. 60680							
F-1 + Hf+4 = Hf+4_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	Unknown	lgK:9.03(0.01SD)	5	CRV	2556
2	25	4	Unknown	dH:-0.8(1SF)	5	CRV	2556

Reaction No. 60681							
2<F-1> + Hf+4 = Hf+4_F-1(2)							

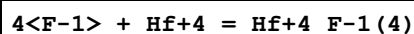
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	Unknown	lgK:16.56(4SF)	5	CRV	2556
2	25	4	Unknown	dH:-1.5(2SF)	5	CRV	2556

Reaction No. 60682



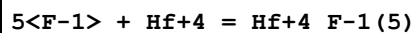
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	4	Unknown	lgK:23.1(3SF)	5	CRV	2556
2	25	4	Unknown	dH:-0.8(1SF)	5	CRV	2556

Reaction No. 60683



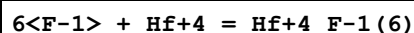
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.3(3SF)	1	EDH	
2	25	4	Unknown	lgK:28.7(3SF)	5	CRV	2556
3	37	0.15	Unknown	lgK:30.2(3SF)	5	CRV	2556
4	25	4	Unknown	dH:-4.5(2SF)	5	CRV	2556

Reaction No. 60684



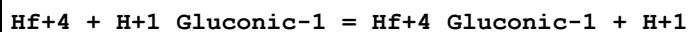
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.(2SF)	5	EES	
2	25	4	Unknown	lgK:34.0(3SF)	5	CRV	2556
3	37	0.15	Unknown	lgK:40.5(3SF)	4	CRV	2556

Reaction No. 60685



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.7(3SF)	1	EDH	
2	25	4	Unknown	lgK:38.0(3SF)	4	CRV	2556

Reaction No. 61465



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO4	lgK:1.49(3SF)	4	MIX	2261

Reaction No. 62970

2<HIMDA-2> + Hf+4 = Hf+4_HIMDA-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:12.0(0.07SD)	6	CRV	552
2	25	0.1	Unknown	dH:-4.4(2SF)	3	CRV	552
3	25	0.5	Unknown	dH:-4.6(2SF)	3	CRV	552
4	25	1	Unknown	dH:-4.7(2SF)	3	CRV	552

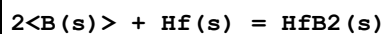
Reaction No. 62971							
Hf+4_OH-1(2)_HIMDA-2 + H+1 = Hf+4_OH-1_HIMDA-2 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0.1	Unknown	lgK:10.85(4SF)	6	CRV	552

Reaction No. 66515							
Ba(s) + Hf(s) + 1.5<O2(g)> = BaHfO3(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:305.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:303.8(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:224.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:176.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:144.4(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-417.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-437.9(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-437.6(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-437.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-437.3(4SF)	4	MTD	6422

Reaction No. 66569							
Ca(s) + Hf(s) + 1.5<O2(g)> = CaO.HfO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:302.8(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:300.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:222.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:174.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:143.6(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-413.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-433.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-431.9(4SF)	4	MTD	6422

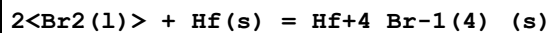
9	227	0	Inf. Dilution	dH:-431.0 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-430.3 (4SF)	4	MTD	6422

Reaction No. 67312



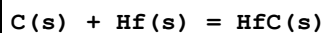
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:58.21 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:57.84 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:43.22 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:34.46 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:28.62 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-79.41 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-80.30 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-80.24 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-80.18 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-80.14 (4SF)	4	MTD	6422

Reaction No. 67313



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:128.7 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:127.9 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:92.89 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:71.36 (4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:57.08 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-175.6 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-183.4 (4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-197.4 (4SF)	2	MTD	6422
9	227	0	Inf. Dilution	dH:-196.5 (4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-195.5 (4SF)	0	MTD	6422

Reaction No. 67314



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:43.56 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:43.29 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:32.36 (4SF)	4	ENS	6422

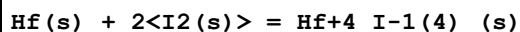
4	227	0	Inf. Dilution	lgK:25.82(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:21.46(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-59.42(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-60.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-59.95(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-59.88(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-59.81(4SF)	4	MTD	6422

Reaction No. 67315							
2<Cl2(g)> + Hf(s) = Hf+4_Cl-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:157.9(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:156.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:113.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:88.00(4SF)	1	ENS	6422
5	315	0	Inf. Dilution	lgK:72.53(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-901.3(4SF) kJ	3	CRV	6330
7	25	0	Inf. Dilution	dG:-215.4(4SF)	3	EFE	6422
8	25	0	Inf. Dilution	dH:-990.4(4SF) kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-236.7(4SF)	2	MTD	6422
10	127	0	Inf. Dilution	dH:-236.0(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-235.3(4SF)	0	MTD	6422
12	315	0	Inf. Dilution	dH:-234.8(4SF)	0	MTD	6422

Reaction No. 67316							
2<F2(g)> + Hf(s) = Hf+4_F-1(4)_(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:320.7(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:318.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:234.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:184.2(4SF)	0	ENS	6422
5	327	0	Inf. Dilution	lgK:150.7(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-437.4(4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dH:-461.4(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-461.0(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-460.4(4SF)	0	MTD	6422

10	327	0	Inf. Dilution	dH:-459.8 (4SF)	0	MTD	6422
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Reaction No. 67317



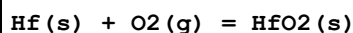
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:86.18 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:85.65 (4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:64.03 (4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:49.48 (4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:38.91 (4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-117.6 (4SF)	3	EFE	6422
7	25	0	Inf. Dilution	dG:-117.580 (6SF)	3	CRV	31291
8	25	0	Inf. Dilution	dH:-118.0 (4SF)	3	MTD	6422
9	25	0	Inf. Dilution	dH:-118.00 (5SF)	3	CRV	31291
10	127	0	Inf. Dilution	dH:-125.5 (4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-145.6 (4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-144.4 (4SF)	0	MTD	6422

Reaction No. 67318



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:60.69 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:60.29 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:44.03 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:34.27 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:27.78 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-82.80 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-89.30 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-89.28 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-89.20 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-89.11 (4SF)	4	MTD	6422

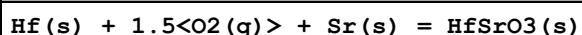
Reaction No. 67319



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:189.4 (4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:190.7 (4SF)	0	ENS	6422

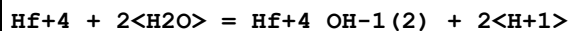
3	27	0	Inf. Dilution	lgK:189.4 (4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:139.6 (4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:109.8 (4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:89.87 (4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-260.1 (4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dH:-273.6 (4SF)	0	MTD	6422
9	127	0	Inf. Dilution	dH:-273.4 (4SF)	0	MTD	6422
10	227	0	Inf. Dilution	dH:-273.1 (4SF)	0	MTD	6422
11	327	0	Inf. Dilution	dH:-272.8 (4SF)	0	MTD	6422

Reaction No. 67320



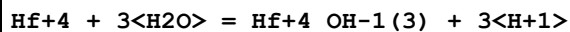
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:298.1 (4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:296.2 (4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:218.5 (4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:171.9 (4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:140.8 (4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-406.7 (4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-426.9 (4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-426.6 (4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-426.2 (4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-425.8 (4SF)	4	MTD	6422

Reaction No. 67447



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.7	Seawater	lgK:-3.49 (3SF)	0	ENS	6496
2	25	0.7	Seawater	dH:18.8 (3SF)	0	ENS	6496

Reaction No. 67448



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-49.1 (3SF)	1	ELC	
2	25	0.7	Seawater	lgK:-7.38 (3SF)	0	ENS	6496
3	25	0.7	Seawater	dH:25.7 (3SF)	0	ENS	6496

Reaction No. 67449							
$\text{Hf}+4 + 4\langle\text{H}_2\text{O}\rangle = \text{Hf}+4_{\text{OH}-1}(4) + 4\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -10.7(3\text{SF})$	4	CRV	32224
2	25	0.7	Seawater	$\lg K: -12.08(4\text{SF})$	0	ENS	6496
3	25	0.7	Seawater	$\text{dH}: 30.4(3\text{SF})$	0	ENS	6496

Reaction No. 67450							
$\text{Hf}+4 + 5\langle\text{H}_2\text{O}\rangle = \text{Hf}+4_{\text{OH}-1}(5) + 5\langle\text{H}+1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -20.3(3\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\lg K: -17.15(4\text{SF})$	0	CRV	32224
3	25	0.7	Seawater	$\lg K: -18.27(4\text{SF})$	0	ENS	6496
4	25	0.7	Seawater	$\text{dH}: 34.0(3\text{SF})$	0	ENS	6496

Reaction No. 71103							
$\text{HfO}(\text{OH})_2(\text{s}) + \text{H}_2\text{O} + 4\langle\text{e}-1\rangle = \text{Hf}(\text{s}) + 4\langle\text{OH}-1\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{E}: -2.50(3\text{SF})$	3	CRV	6330

Reaction No. 71104							
$\text{HfO}+2 + 2\langle\text{H}+1\rangle + 4\langle\text{e}-1\rangle = \text{Hf}(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{E}: -1.724(4\text{SF})$	5	CRV	6330

Reaction No. 71659							
$\text{Hf}+4_{\text{OH}-1} + \text{H}+1 + 4\langle\text{e}-1\rangle = \text{Hf}(\text{s}) + \text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: -104.6(4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$\text{E}: -1.55(3\text{SF})$	0	CRV	7761

Reaction No. 71660							
$\text{Hf}+4(4)_{\text{OH}-1}(8) + 8\langle\text{H}+1\rangle + 16\langle\text{e}-1\rangle = 4\langle\text{Hf}(\text{s})\rangle + 8\langle\text{H}_2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\text{E}: -1.57(3\text{SF})$	3	CRV	7761

Reaction No. 71661							
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Hf+4_OH-1(5) + 4<e-1> = Hf(s) + 5<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.31(3SF)	0	CRV	7761
Note (1): Low wgt for inter-reaction consistency							

Reaction No. 71662							
HfO2(am.,s) + 2<H2O> + 4<e-1> = Hf(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-159.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.37(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:776.6(4SF)kJ	2	CRV	7761

Reaction No. 71663							
HfO2(s) + 2<H2O> + 4<e-1> = Hf(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-162.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.419(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:796.7(4SF)kJ	2	CRV	7761

Reaction No. 72282							
HfO2(s) + 4<H+1> = Hf+4 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.5(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.2(1.SD)	0	CRV	6478
3	25	0	Inf. Dilution	lgK:1.16(3SF)	0	CRV	32224

Reaction No. 73475							
6<OH-1> + Hf+4 = Hf+4_OH-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.2(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:51.2(0.2SD)	4	MSL	14521

Reaction No. 73476							
4<OH-1> + Hf+4 = Hf+4_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.8(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:44.8(3SF)	3	MSL	14521

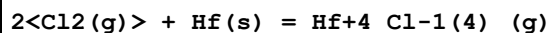
Note (2): <44.8

Reaction No. 73477



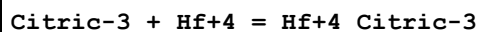
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-55.1(0.7SD)	4	MSL	14521

Reaction No. 75624



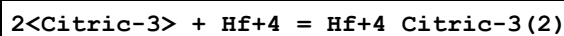
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-884.5(4SF)kJ	3	CRV	6330

Reaction No. 75888



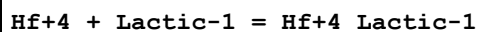
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.269(4SF)	1	EOR	
2	25	2	NaClO4	lgK:5.33(3SF)	5	MDS	13484

Reaction No. 75889



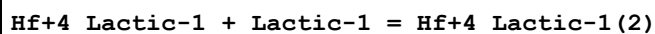
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.78(4SF)	1	EOR	
2	25	2	NaClO4	lgK:9.11(3SF)	5	MDS	13484

Reaction No. 75920



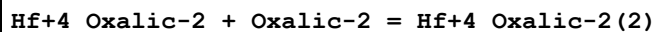
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:5.61(3SF)	5	MDS	13484

Reaction No. 75921



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:5.43(3SF)	5	MDS	13484

Reaction No. 75938



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	2	NaClO4	lgK:8.66(3SF)	5	MDS	11516

Note (1): Navratil O et al., Collec. Czech. Chem. Commun., 1971, 36, 1659							
2	25	0	Inf. Dilution	lgK:8.5 (2SF)	1	ESC	

Reaction No. 76345							
Hf+4_Tartaric-2 + Tartaric-2 = Hf+4_Tartaric-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:4.00 (3SF)	5	MDS	13484

Reaction No. 76360							
Hf+4 + Glycolic-1 = Hf+4_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:4.94 (3SF)	5	MDS	13484

Reaction No. 76361							
Hf+4_Glycolic-1 + Glycolic-1 = Hf+4_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	2	NaClO4	lgK:4.91 (3SF)	5	MDS	13484

Reaction No. 78023							
3<OH-1> + Hf+4 = Hf+4_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.08 (4SF)	1	ELC	

Reaction No. 78107							
Gly-1 + Hf+4 + H+1 = Hf+4_H+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.3 (4SF)	1	ELC	

Reaction No. 78329							
Acetic-1 + 2<OH-1> + Hf+4 = Hf+4_Acetic-1_OH-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.7 (4SF)	1	ELC	

Reaction No. 78370							
2<Acetic-1> + 3<OH-1> + Hf+4 = Hf+4_Acetic-1(2)_OH-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.062 (4SF)	1	ELC	

Reaction No. 79038							
$4\text{<I-1>} + \text{Hf+4} = \text{Hf+4_I-1(4)_(s)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-54.98 (4SF)	1	ELC	

Reaction No. 79159							
$8\text{<OH-1>} + 4\text{<Hf+4>} = \text{Hf+4(4)_OH-1(8)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:117.4 (4SF)	1	ELC	

Reaction No. 80205							
$\text{Hf(s)} + 2\text{<I2(s)>} = \text{Hf+4_I-1(4)_(g)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-98.930 (5SF)	3	CRV	31291
2	25	0	Inf. Dilution	dH:-87.000 (5SF)	3	CRV	31291

Reaction No. 81448							
$\text{Hf+4_F-1(4)_(s)} = \text{Hf+4} + 4\text{<F-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.2 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-25.746 (5SF)	0	CRV	32224

Reaction No. 81449							
$\text{Hf+4_Cl-1(4)_(s)} = \text{Hf+4} + 4\text{<Cl-1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.9 (3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:32.095 (5SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination

4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MCE	Cation exchange
MCL	Calorimetry
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MIX	Ion exchange
MKN	Rate of reaction
MLC	Ligand competition
MPH	pH method, not specified
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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